

ASYMMETRIC KEY ENCRYPTION

Cryptography

CRYPTOGRAPHIC ALGORITHMS - DRAWBACK OF SYMMETRIC ENCRYPTION

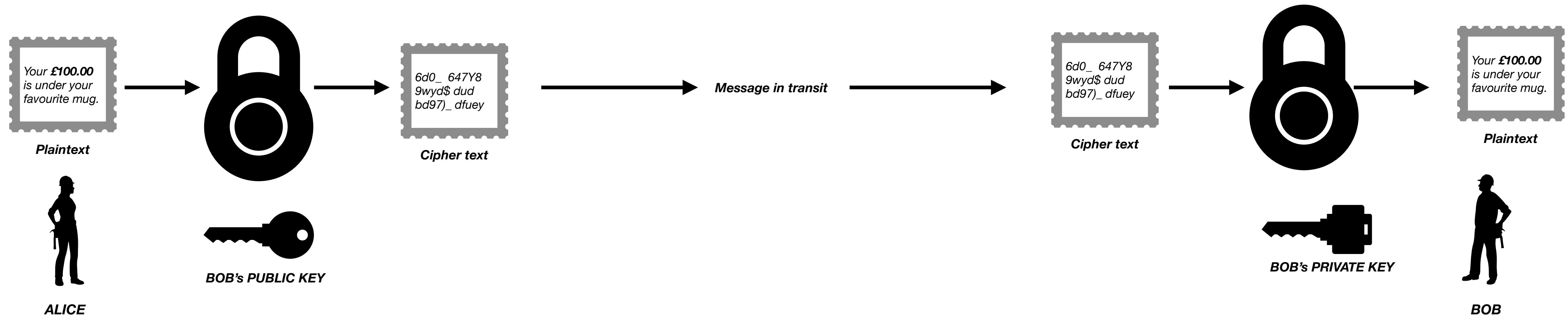
- Symmetric key cryptography is not scalable. There should be a key for each pair of sender and receiver. This may work for small groups but becomes infeasible as the group size increases.
- Key distribution is not easy. There is no easy and secure way for the sender and receiver to exchange the secret key. (The confidentiality of secret keys needs to be preserved during key distribution: this is harder to achieve than integrity, which is all that is required in the case of public keys.)
- Can't easily be used for digital signatures.

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CRYPTOGRAPHIC ALGORITHMS - ASYMMETRIC ENCRYPTION

Asymmetric encryption also referred to as public key cryptography. It uses two keys instead of only one. The keys are mathematically related and are known as the **public key** and the **private key**. The **public key is known to everyone** and can be freely distributed, while the **private key is known only to the individual to whom it belongs**.

When Alice wants to send a secure message to Bob, she uses Bob's public key to encrypt the message. Bob then uses his private key to decrypt it.



PRINCIPLES

Public Key: Public key are not kept secret and they can be shared with anyone.

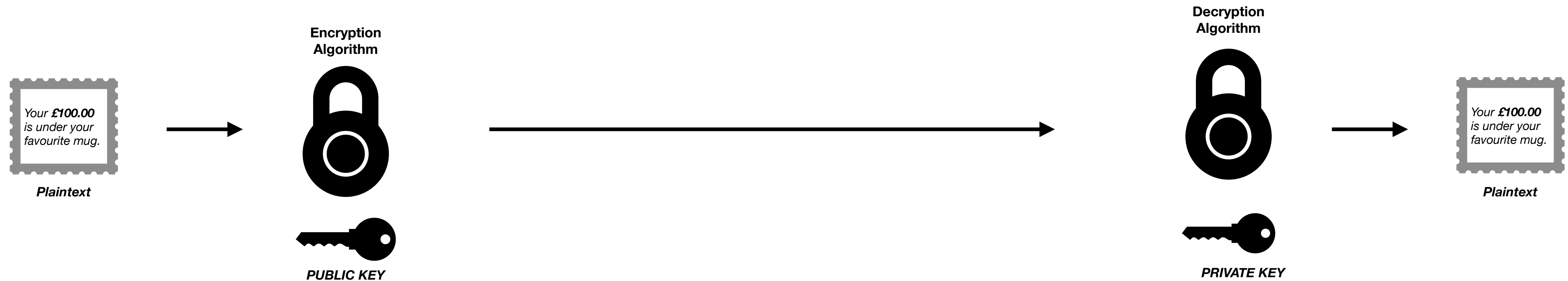
Private Key: Private key should be kept secret and not be shared with anyone.

Key Pair: Asymmetric relies on both public and private keys unlike symmetric encryption.

Both directions: Asymmetric works both directions, a file encrypted with private key can be decrypted by public key.

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ADVANTAGES

- Wide publication of public keys is possible, for instance with **public key certificates** that bind an identifier to a public key thanks to a trusted third party (called **Certification Authority**).
- Can provide digital signatures that can be repudiated

DISADVANTAGES

- Slow processing: at least 100 times slower than symmetric cryptography

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RIVEST SHAMIR ADLEMAN (RSA)

The RSA algorithm is an asymmetric cryptography algorithm published in 1977. It uses a public key and a private key (i.e., two different, mathematically linked keys).

RSA derives its security from the difficulty of factoring large integers that are the product of two large prime numbers. Multiplying these two numbers is easy, but determining the original prime numbers from the total -- or ***factoring*** -- is considered infeasible due to the time it would take using even today's supercomputers.

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The RSA algorithm is an asymmetric cryptography algorithm published in 1977. It uses a public key and a private key (i.e., two different, mathematically linked keys).

The RSA algorithm multiplies two large prime numbers (a prime number is a number divisible only by itself and 1), p and q , to compute their product ($n = pq$). Next, a number e is chosen that is less than n and a prime factor to $(p - 1)(q - 1)$. Another number d is determined so that $(ed - 1)$ is divisible by $(p - 1)(q - 1)$. The values of e and d are the public and private exponents. The public key is the pair (n, e) while the private key is (n, d) . The numbers p and q can be discarded.

RSA algorithm using very small numbers is as follows:

1. Select two prime numbers, p and q (in this example, $p = 7$ and $q = 19$).
2. Multiply p and q together to create n ($7 * 19 = 133$).
3. Calculate m as $(p - 1 * q - 1)$ ($[7 - 1] => ([19 - 1])$ or $(6 * 18) = 108$).
4. Find a number e so that it and m have no common positive divisor other than 1 ($e = 5$).
5. Find a number d so that $d = (1 + n * m)/e$ or $([1 + 133 * 108]/5$ or $14,364/5 = 2,875$).

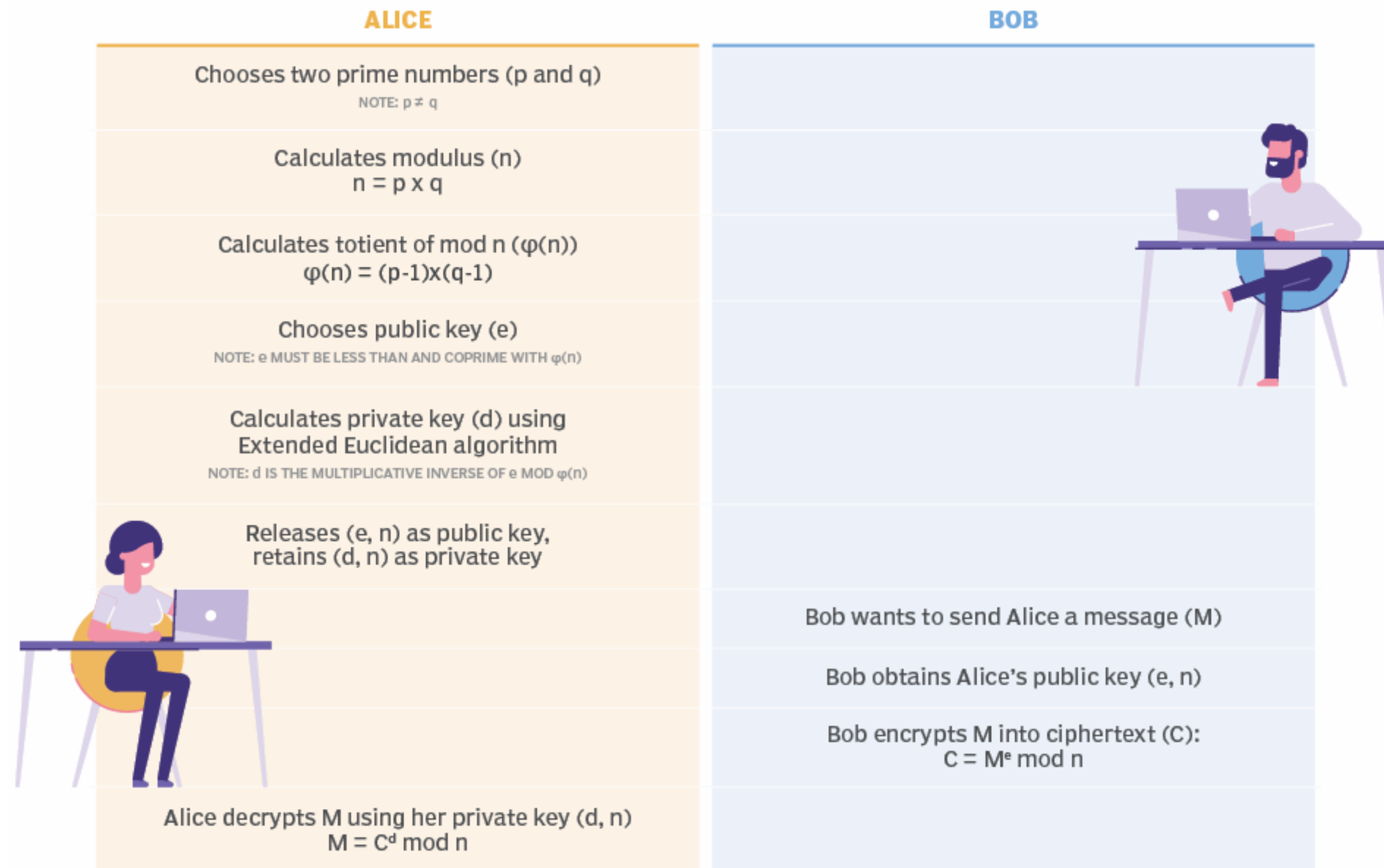
For this example, the public key n is **133** and e is **5**, while for the private key n is **133** and d is **2873**.

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CRYPTOGRAPHIC ALGORITHMS - ASYMMETRIC ENCRYPTION RIVEST SHAMIR ADLEMAN (RSA)

Alice's Setup:

- Chooses two prime numbers.
- Calculates the product $n = pq$.
- Calculates $m = (p - 1)(q - 1)$.
- Chooses numbers e and d so that ed has a remainder of 1 when divided by m .
- Publishes her public key (n, e) .

Bob encrypts a message M for Alice:

- Finds Alice's public key (n, e) .
- Finds the remainder C when M^e is divided by n .
- Sends ciphertext C to Alice.

Alice receives and decrypts ciphertext C :

- Uses her private key (n, d) .
- Finds remainder R when C^d is divided by n .
- R matches the message M that Bob wanted to send to Alice!

Alice's Setup:

- $p = 11$ and $q = 3$.
- $n = pq = 11 \times 3 = 33$.
- $m = (p - 1)(q - 1) = 10 \times 2 = 20$.
- If $e = 3$ and $d = 7$, then $ed = 21$ has a remainder of 1 when divided by $m = 20$.
- Publish $(n, e) = (33, 3)$.

Bob encrypts message $M = 14$:

$$C = m^e \bmod n$$

- $(n, e) = (33, 3)$.
- When $14^3 = 2744$ is divided by 33, the remainder is $C = 5$.
- Sends ciphertext $C = 5$ to Alice.

Alice decrypts ciphertext $C = 5$:

$$m = c^d \bmod n$$

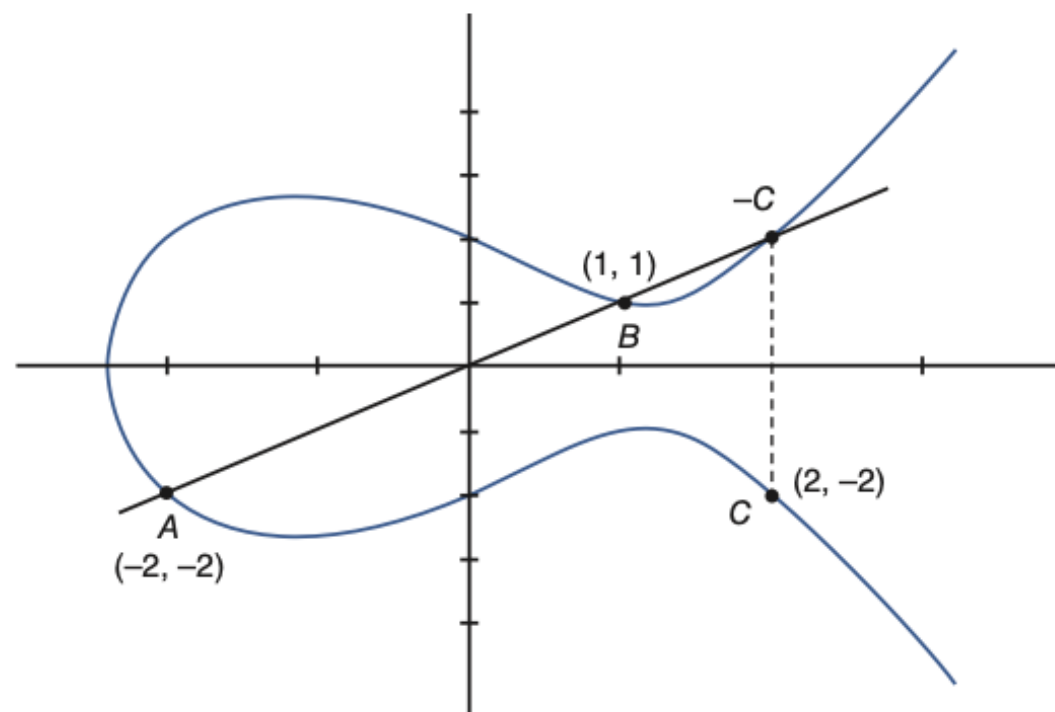
- $(n, d) = (33, 7)$.
- When $5^7 = 78125$ is divided by 33, the remainder is $R = 14$.
- $R = 14 = M$, the original message from Bob!

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CRYPTOGRAPHIC ALGORITHMS - ASYMMETRIC ENCRYPTION

ELLIPTIC CURVE CRYPTOGRAPHY (ECC)

- The basis of RSA asymmetric encryption security is factoring, or the *prime numbers* that make up a *value*.
- The strength of the RSA algorithm relies on the fact that large numbers are difficult/take long time to factorise. Hence, the algorithm depends on large prime numbers (e.g. 512/1024 bits) to produce a hard-to-factorise large number.
- Conventional techniques are inefficient to generate/test for prime numbers.
- Computers have become faster and more powerful so the ability to “crack” RSA asymmetric encryption by computing the factoring has grown.

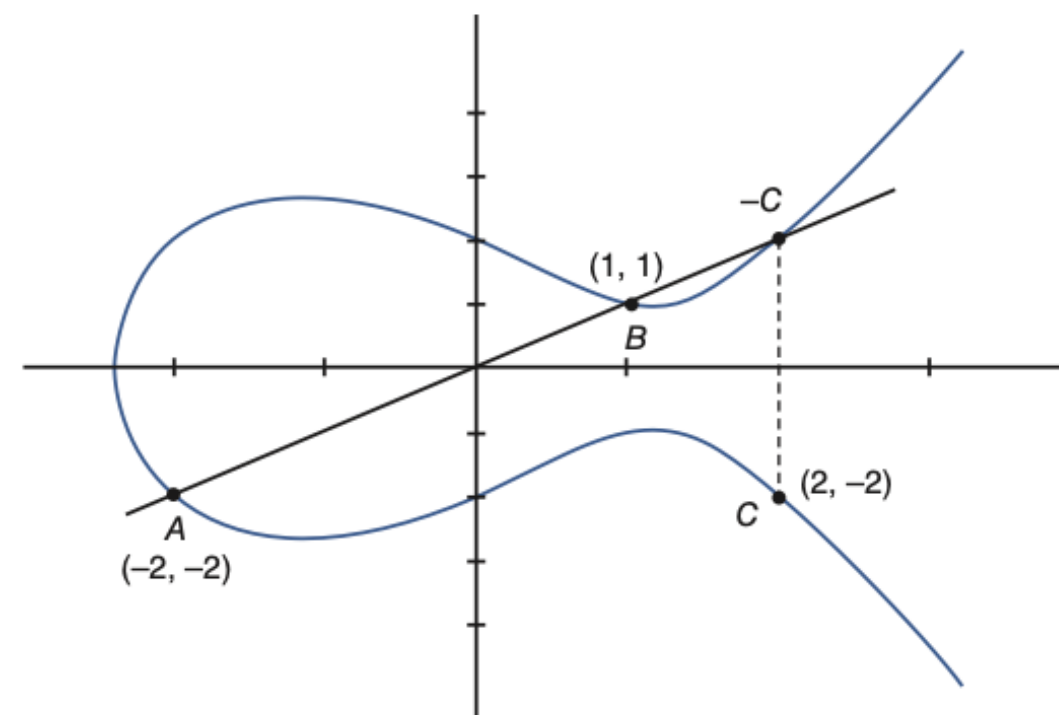


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ELLIPTIC CURVE CRYPTOGRAPHY (ECC)

- Obscure (and esoteric) branch of mathematics called elliptic curves has been explored as a prevention.
- An elliptic curve is a set of points that satisfy a specific mathematical equation. Elliptic curves paved the way for a different form of asymmetric encryption not based on factoring.
- Instead of using large prime numbers as with RSA, **elliptic curve cryptography (ECC)** uses sloping curves. An elliptic curve is a function drawn on an X-Y axis as a gently curved line. By adding the values of two points on the curve, a third point on the curve can be derived, of which the inverse is used.
- With ECC, users share one elliptic curve and one point on the curve. One user chooses a secret random number and computes a public key based on a point on the curve; the other user does the same. They can now exchange messages because the shared public keys can generate a private key on an elliptic curve.



An elliptic curve for current ECC purposes is a plane curve over a finite field which is made up of the points satisfying the equation:

$$y^2 = x^3 + ax + b$$

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ECC VS RSA

- There is no linear relationship between the sizes of ECC keys and RSA keys.
- In contrast to RSA, ECC bases its approach to public key cryptographic systems on how elliptic curves are structured algebraically over finite fields.
- ECC creates keys that are more difficult, mathematically, to crack. For this reason, ECC is considered to be the next generation implementation of public key cryptography and more secure than RSA.
- ECC key generation and signing are substantially quicker than for RSA, and also that ECC uses less memory than does RSA.
- Unlike in RSA, where both are integers, in ECC the private and public keys are not equally exchangeable. Instead, in ECC the public key is a point on the curve, while the private key is still an integer.

RSA Key length	ECC Key length
1024	160
2048	224
3072	256
7680	384
15360	521

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ECC VS RSA

ADVANTAGES

- ECC features smaller ciphertexts, keys, and signatures, and faster generation of keys and signatures. Its decryption and encryption speeds are moderately fast.
- ECC enables lower latency than inverse throughout by computing signatures in two stages.
- ECC features strong protocols for authenticated key exchange and support for the tech is strong.

DISADVANTAGES

- The main disadvantage of ECC is that it isn't easy to securely implement. Compared to RSA, which is much simpler on both the verification and encryption sides, ECC is a steeper learning curve and a bit slower for accumulating actionable results.

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- An Elliptic Curve Digital Signature Algorithm (ECDSA) uses ECC keys to ensure each user is unique and every transaction is secure.
- ECC is among the most commonly used implementation techniques for digital signatures in cryptocurrencies. Both Bitcoin and Ethereum apply the Elliptic Curve Digital Signature Algorithm (ECDSA) specifically in signing transactions.

Cryptography

SYMMETRIC VS ASYMMETRIC ENCRYPTION

Feature	Symmetric	Asymmetric
Speed of execution	Conventional cryptography is fast. Software implementation on current PCs can perform encryption at the rate of few MBps.	Public key cryptography is much slower. About 3 orders of magnitude slower.
Key distribution	Key distribution is difficult. There is no easy and secure way by which sender and receiver can exchange the secret key, since confidentiality of keys cannot be compromised.	Key distribution is easier because the public keys can be freely distributed. Integrity of public keys should not be compromised but confidentiality is not needed.
Non-repudiability	Since the sender and receiver share the same key, non-repudiability is compromised	Non-repudiability is preserved